

SAFETY DATA SHEET

FYFANON 50% W/W EC



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1. PRODUCT AND COMPANY IDENTIFICATION

Product name : FYFANON 50% W/W EC

Other means of identification : MALATHION 526 g/L EC

Recommended use of the chemical and restrictions on use

Recommended use : Insecticide

Restrictions on use : Use as recommended by the label.

Manufacturer or supplier's details

Company : FMC Taiwan Co. Ltd.

Address : 3F, No. 179, Fuhuiyuan Blvd.,
Xitun Dist., Taichung City, 40758
Taiwan

Telephone : +886 4 2254 2785

Telefax : +886 4 2254 2831

E-mail address : SDS-Info@fmc.com

Emergency telephone : For leak, fire, spill or accident emergencies, call:
00801-14-8954 (CHEMTREC)

Medical emergency:
00886 2 28757525 (in case of chemical spill, fire or poisoning
accident)

2. HAZARDS IDENTIFICATION

GHS Classification

Flammable liquids : Category 4

Acute toxicity (Oral) : Category 5

Acute toxicity (Inhalation) : Category 5

Acute toxicity (Dermal) : Category 5

Skin corrosion/irritation : Category 3

Serious eye damage/eye irri- : Category 1

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tation

Skin sensitization : Category 1

Specific target organ toxicity - single exposure : Category 3 (Central nervous system)

Aspiration hazard : Category 1

Short-term (acute) aquatic hazard : Category 1

Long-term (chronic) aquatic hazard : Category 1

GHS label elements

Hazard pictograms :



Signal Word : DANGER

Hazard Statements :

H227 Combustible liquid.
H303 + H313 + H333 May be harmful if swallowed, in contact with skin or if inhaled.
H304 May be fatal if swallowed and enters airways.
H316 Causes mild skin irritation.
H317 May cause an allergic skin reaction.
H318 Causes serious eye damage.
H336 May cause drowsiness or dizziness.
H410 Very toxic to aquatic life with long lasting effects.

Precautionary Statements :

Prevention:

P210 Keep away from heat/ sparks/ open flames/ hot surfaces. No smoking.
P261 Avoid breathing mist or vapors.
P271 Use only outdoors or in a well-ventilated area.
P272 Contaminated work clothing should not be allowed out of the workplace.
P273 Avoid release to the environment.
P280 Wear protective gloves/ eye protection/ face protection.

Response:

P301 + P310 IF SWALLOWED: Immediately call a POISON CENTER/ doctor.
P302 + P352 IF ON SKIN: Wash with plenty of water.
P304 + P340 + P312 IF INHALED: Remove person to fresh air and keep comfortable for breathing. Call a POISON CENTER/ doctor if you feel unwell.
P305 + P351 + P338 + P310 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present

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and easy to do. Continue rinsing. Immediately call a POISON CENTER/ doctor.
P312 Call a POISON CENTER/ doctor if you feel unwell.
P331 Do NOT induce vomiting.
P333 + P313 If skin irritation or rash occurs: Get medical advice/ attention.
P362 + P364 Take off contaminated clothing and wash it before reuse.
P370 + P378 In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish.
P391 Collect spillage.

Storage:

P403 + P233 Store in a well-ventilated place. Keep container tightly closed.
P403 + P235 Store in a well-ventilated place. Keep cool.
P405 Store locked up.

Disposal:

P501 Dispose of contents/ container to an approved waste disposal plant.

Other hazards which do not result in classification

None known.

3. COMPOSITION/INFORMATION ON INGREDIENTS

Substance / Mixture : Mixture

Components

Hazardous ingredients	CAS-No.	Concentration (% w/w)
malathion (ISO) [containing $\leq 0,03$ % isomalathion]	121-75-5	≥ 30 -< 55
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified	64742-94-5	≥ 30 -< 50
acetic anhydride	108-24-7	≥ 1 -< 3
calcium bis(dodecylbenzenesulphonate), branched	70528-83-5	≥ 1 -< 2.5

4. FIRST AID MEASURES

General advice : Move out of dangerous area.
Consult a physician.
Show this material safety data sheet to the doctor in attendance.
Symptoms of poisoning may appear several hours later.
Do not leave the victim unattended.

First aid measures for different exposure routes

If inhaled : Move to fresh air.
Consult a physician after significant exposure.

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- If unconscious, place in recovery position and seek medical advice.
If experiencing any discomfort, immediately remove from exposure. Light cases: Keep person under surveillance. Get medical attention immediately if symptoms develop. Serious cases: Get medical attention immediately or call for an ambulance.
- In case of skin contact : Take off all contaminated clothing immediately.
Wash contaminated clothing before re-use.
Wash off immediately with plenty of water for at least 15 minutes.
Get medical attention immediately if irritation develops and persists.
- In case of eye contact : Small amounts splashed into eyes can cause irreversible tissue damage and blindness.
In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.
Continue rinsing eyes during transport to hospital.
Remove contact lenses.
Protect unharmed eye.
Keep eye wide open while rinsing.
If eye irritation persists, consult a specialist.
- If swallowed : Keep respiratory tract clear.
Do NOT induce vomiting.
Do not give milk or alcoholic beverages.
Never give anything by mouth to an unconscious person.
If symptoms persist, call a physician.
Take victim immediately to hospital.
- Most important symptoms and effects, both acute and delayed : May be harmful if swallowed, in contact with skin or if inhaled.
May be fatal if swallowed and enters airways.
Causes mild skin irritation.
May cause an allergic skin reaction.
Causes serious eye damage.
May cause drowsiness or dizziness.
- Protection of first-aiders : First Aid responders should pay attention to self-protection and use the recommended protective clothing
Avoid inhalation, ingestion and contact with skin and eyes.
If potential for exposure exists refer to Section 8 for specific personal protective equipment.
- Notes to physician : Treat symptomatically.

5. FIRE-FIGHTING MEASURES

- Suitable extinguishing media : Dry chemical, CO₂, water spray or regular foam.
Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

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|--|---|
| Unsuitable extinguishing media | : High volume water jet
Do not spread spilled material with high-pressure water streams. |
| Specific hazards during fire fighting | : Do not allow run-off from fire fighting to enter drains or water courses. |
| Hazardous combustion products | : Fire may produce irritating, corrosive and/or toxic gases.
phosphorus oxides
Carbon oxides
Sulfur oxides |
| Specific extinguishing methods | : Collect contaminated fire extinguishing water separately. This must not be discharged into drains.
Fire residues and contaminated fire extinguishing water must be disposed of in accordance with local regulations.
For safety reasons in case of fire, cans should be stored separately in closed containments.
Use a water spray to cool fully closed containers. |
| Special protective equipment for fire-fighters | : Firefighters should wear protective clothing and self-contained breathing apparatus. |

6. ACCIDENTAL RELEASE MEASURES

- | | |
|---|---|
| Personal precautions, protective equipment and emergency procedures | : Use personal protective equipment.
Ensure adequate ventilation.
Never return spills in original containers for re-use.
Mark the contaminated area with signs and prevent access to unauthorized personnel.
Only qualified personnel equipped with suitable protective equipment may intervene.
For disposal considerations see section 13. |
| Environmental precautions | : Prevent product from entering drains.
Prevent further leakage or spillage if safe to do so.
If the product contaminates rivers and lakes or drains inform respective authorities. |
| Methods and materials for containment and cleaning up | : Contain spillage, and then collect with non-combustible absorbent material, (e.g. sand, earth, diatomaceous earth, vermiculite) and place in container for disposal according to local / national regulations (see section 13).
Keep in suitable, closed containers for disposal. |

7. HANDLING AND STORAGE

Handling

- | | |
|---|--|
| Advice on protection against fire and explosion | : Do not spray on a naked flame or any incandescent material.
Keep away from open flames, hot surfaces and sources of ignition. |
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Advice on safe handling : Avoid formation of aerosol.
Do not breathe vapors/dust.
Avoid exposure - obtain special instructions before use.
Avoid contact with skin and eyes.
For personal protection see section 8.
Smoking, eating and drinking should be prohibited in the application area.
Provide sufficient air exchange and/or exhaust in work rooms.
To avoid spills during handling keep bottle on a metal tray.
Dispose of rinse water in accordance with local and national regulations.
Persons susceptible to skin sensitization problems or asthma, allergies, chronic or recurrent respiratory disease should not be employed in any process in which this mixture is being used.

Storage

Conditions for safe storage : No smoking.
Keep container tightly closed in a dry and well-ventilated place.
Containers which are opened must be carefully resealed and kept upright to prevent leakage.
Observe label precautions.
Electrical installations / working materials must comply with the technological safety standards.

Further information on storage stability : No decomposition if stored and applied as directed.

8. EXPOSURE CONTROLS/PERSONAL PROTECTION**Ingredients with workplace control parameters**

Components	CAS-No.	Value type (Form of exposure)	Control parameters / Permissible concentration	Basis
malathion (ISO) [containing ≤ 0,03 % isomalathion]	121-75-5	TWA	10 mg/m3	TW OEL
		Further information: Skin		
		STEL	15 mg/m3	TW OEL
		Further information: Skin		
Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified	64742-94-5	TWA	200 mg/m3 (total hydrocarbon vapor)	ACGIH
		TWA (Inhalable fraction and vapor)	1 mg/m3	ACGIH
acetic anhydride	108-24-7	TWA	5 ppm 21 mg/m3	TW OEL
		STEL	10 ppm 31.5 mg/m3	TW OEL
		TWA	1 ppm	ACGIH

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		STEL	3 ppm	ACGIH
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Biological occupational exposure limits

Contains no substances with biological exposure indices.

Personal protective equipment

Respiratory protection : In case of mist, spray or aerosol exposure wear suitable personal respiratory protection and protective suit.

Hand protection
Material : Wear chemical resistant gloves, such as barrier laminate, butyl rubber or nitrile rubber.

Remarks : The suitability for a specific workplace should be discussed with the producers of the protective gloves.

Eye protection : Eye wash bottle with pure water
Tightly fitting safety goggles
Wear face-shield and protective suit for abnormal processing problems.

Skin and body protection : Impervious clothing
Choose body protection according to the amount and concentration of the dangerous substance at the work place.

Protective measures : In the context of professional plant protection use as recommended, the end user must refer to the label and the instructions for use.

Ensure that eye flushing systems and safety showers are located close to the working place.
Plan first aid action before beginning work with this product.
Always have on hand a first-aid kit, together with proper instructions.
Wear suitable protective equipment.

Hygiene measures : When using do not eat or drink.
When using do not smoke.
Wash hands before breaks and at the end of workday.

9. PHYSICAL AND CHEMICAL PROPERTIES

Physical state : liquid

Color : yellowish-brown

Odor : Aromatic

Odor Threshold : No data available

pH : acidic

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Melting point/freezing point : < 0 °C

Initial boiling point and boiling range : No data available

Flash point : 71 °C

Method: Pensky-Martens closed cup

Upper explosion limit / Upper flammability limit : No data available

Lower explosion limit / Lower flammability limit : No data available

Vapor pressure : No data available

Relative vapor density : No data available

Relative density : No data available

Density : 1,052 g/l

Bulk density : No data available

Solubility(ies)
Water solubility : Miscible

Solubility in other solvents : 250 g/l (20 °C)
Active ingredient
Solvent: ethyl acetate

57 - 67 g/l (20 °C)
Active ingredient
Solvent: Heptane

148.2 mg/l (25 °C)
Active ingredient
Solvent: water

Partition coefficient: n-octanol/water : No data available

Autoignition temperature : No data available

Decomposition temperature : No data available

Viscosity
Viscosity, dynamic : No data available

Viscosity, kinematic : No data available

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Explosive properties	: Not explosive
Oxidizing properties	: Non-oxidizing
Particle size	: No data available

10. STABILITY AND REACTIVITY

Reactivity	: No decomposition if stored and applied as directed.
Chemical stability	: Malathion will decompose rapidly when heated to temperatures above 140°C, significantly increasing the risk of explosion. Direct local heating such as electric heating or by steam must be avoided. The decomposition is dependent on time as well as temperature due to self-accelerating exothermic and autocatalytic reactions. The reactions involve rearrangements and polymerisation releasing volatile malodorous and inflammable compounds such as dimethyl sulphide and methyl mercaptan. Malathion will decompose rapidly when heated to temperatures above 100°C, significantly increasing the risk of explosion. Direct local heating such as electric heating or by steam must be avoided.
Possibility of hazardous reactions	: No decomposition if stored and applied as directed. Vapors may form explosive mixture with air. An acid-base neutralisation reaction can be hazardous because of heat release.
Conditions to avoid	: Heat, flames and sparks. Avoid formation of aerosol. Avoid extreme temperatures. Heating of the mixture may evolve harmful and irritant vapours.
Incompatible materials	: Avoid strong acids, bases, and oxidizers. Strong alkalis, amines and strong oxidising compounds. The product can corrode metals (but does not meet the criteria for classification).
Hazardous decomposition products	: phosphorus oxides Carbon oxides Sulfur oxides

11. TOXICOLOGICAL INFORMATION

Symptoms of Overexposure	: None known.
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Acute toxicity

May be harmful if swallowed, in contact with skin or if inhaled.

May be harmful if swallowed, in contact with skin or if inhaled.

Product:

Acute oral toxicity	: LD50 (Rat): > 2,000 mg/kg Method: OECD Test Guideline 425 Remarks: Based on data from similar materials
Acute inhalation toxicity	: LC50 (Rat): > 4.9 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: OECD Test Guideline 403 Assessment: The component/mixture is minimally toxic after short term inhalation. Remarks: Based on data from similar materials
Acute dermal toxicity	: LD50 (Rat): > 4,000 mg/kg Method: OECD Test Guideline 402 Remarks: Based on data from similar materials

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Acute oral toxicity	: LD50 (Rat): 1,857 mg/kg Method: OECD Test Guideline 401 LD50 (Rat, female): 1,608 - 2,550 mg/kg Method: OECD Test Guideline 401 Symptoms: Tremors, hypoactivity GLP: yes
Acute inhalation toxicity	: LC50 (Rat): > 5.2 mg/l Exposure time: 4 h Test atmosphere: dust/mist Method: EPA OPP 81 - 3 GLP: yes Remarks: no mortality
Acute dermal toxicity	: LD50 (Rat): > 2,000 mg/kg Method: US EPA Test Guideline OPP 81-2 GLP: yes Assessment: The component/mixture is minimally toxic after single contact with skin. LD50 (Rat): > 2,000 mg/kg Method: OECD Test Guideline 402 GLP: yes Assessment: The component/mixture is minimally toxic after single contact with skin.

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Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Acute oral toxicity	: LD50 (Rat, male and female): > 5,000 mg/kg Method: OECD Test Guideline 401 Remarks: Based on data from similar materials
Acute inhalation toxicity	: LC50 (Rat): > 4.688 mg/l Exposure time: 4 h Test atmosphere: vapor Assessment: The substance or mixture has no acute inhalation toxicity
Acute dermal toxicity	: LD50 (Rabbit): > 2,000 mg/kg Method: OECD Test Guideline 402 Assessment: The substance or mixture has no acute dermal toxicity

acetic anhydride:

Acute oral toxicity	: LD50 (Rat, male and female): 630 mg/kg
Acute inhalation toxicity	: LC50 (Rat, male and female): 1.670 mg/l Exposure time: 6 h Test atmosphere: vapor

calcium bis(dodecylbenzenesulphonate), branched:

Acute oral toxicity	: Acute toxicity estimate: 3,333 mg/kg
Acute dermal toxicity	: Acute toxicity estimate: 1,470 mg/kg

Skin corrosion/irritation

Causes skin irritation.
Causes mild skin irritation.

Product:

Assessment	: Not classified as irritant
Method	: OECD Test Guideline 404
Result	: Mild skin irritation
Remarks	: Based on data from similar materials

Components:**malathion (ISO) [containing ≤ 0,03 % isomalathion]:**

Species	: Rabbit
Method	: US EPA Test Guideline OPP 81-5
Result	: No skin irritation
GLP	: yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rabbit
Assessment	: Repeated exposure may cause skin dryness or cracking.

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Result	: No skin irritation
Remarks	: Minimal effects that do not meet the threshold for classification. Based on data from similar materials

acetic anhydride:

Result	: Corrosive after 3 minutes to 1 hour of exposure
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calcium bis(dodecylbenzenesulphonate), branched:

Result	: Skin irritation
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Serious eye damage/eye irritation

Causes serious eye damage.
Causes serious eye damage.

Product:

Result	: Eye irritation
Assessment	: Risk of serious damage to eyes.
Method	: OECD Test Guideline 405
Remarks	: Based on data from similar materials

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Species	: Rabbit
Result	: No eye irritation
Method	: EPA OPP 81-4
GLP	: yes

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rabbit
Assessment	: No eye irritation
Remarks	: Minimal effects that do not meet the threshold for classification. Based on data from similar materials

acetic anhydride:

Species	: Rat
Result	: slight irritation

calcium bis(dodecylbenzenesulphonate), branched:

Result	: Irreversible effects on the eye
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Respiratory or skin sensitization

Skin sensitization

Not classified based on available information.

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Skin sensitization

May cause an allergic skin reaction.

Respiratory sensitization

Not classified based on available information.

Respiratory sensitization

Based on available data, the classification criteria are not met.

Product:

Assessment	:	The product is a skin sensitizer, sub-category 1B.
Method	:	OECD Test Guideline 429
Result	:	May cause sensitization by skin contact.
Remarks	:	Based on data from a similar product.

Remarks	:	Causes sensitization.
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Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Routes of exposure	:	Dermal
Species	:	Guinea pig
Method	:	US EPA Test Guideline OPP 81-6
Result	:	Does not cause skin sensitization.
GLP	:	yes

Test Type	:	Local lymph node assay (LLNA)
Routes of exposure	:	Dermal
Species	:	mice
Method	:	OECD Test Guideline 429
Result	:	Does not cause skin sensitization.
GLP	:	yes

Test Type	:	Maximization Test
Routes of exposure	:	Dermal
Species	:	Guinea pig
Method	:	OECD Test Guideline 406
Result	:	May cause sensitization by skin contact.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Test Type	:	Maximization Test
Species	:	Guinea pig
Result	:	Not a skin sensitizer.
Remarks	:	Based on data from similar materials

Chronic toxicity

Germ cell mutagenicity

Not classified based on available information.

Based on available data, the classification criteria are not met.

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Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

Genotoxicity in vitro : Test Type: Ames test
Result: negative

Test Type: In vitro mammalian cell gene mutation test
Result: positive
Remarks: Based on data from similar materials

Test Type: unscheduled DNA synthesis assay
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: chromosome aberration assay
Species: Rat
Result: negative
Remarks: Based on data from similar materials

Test Type: unscheduled DNA synthesis assay
Species: Rat
Result: negative
Remarks: Based on data from similar materials

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Genotoxicity in vitro : Test Type: reverse mutation assay
Method: OECD Test Guideline 471
Result: negative
Remarks: Based on data from similar materials

Genotoxicity in vivo : Test Type: Bone marrow chromosome aberration.
Species: Rat
Application Route: inhalation (vapor)
Result: negative

acetic anhydride:

Genotoxicity in vitro : Test Type: reverse mutation assay
Result: negative

Test Type: Chromosome aberration test in vitro
Result: negative
Remarks: Based on data from similar materials

Test Type: In vitro mammalian cell gene mutation test
Result: Conflicting results have been seen in different studies.

Genotoxicity in vivo : Test Type: Micronucleus test
Species: Rat (male and female)
Application Route: inhalation (vapor)
Result: negative

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Carcinogenicity

Suspected of causing cancer.

Based on available data, the classification criteria are not met.

Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

Species	: Rat
Application Route	: Ingestion
Exposure time	: 24 month(s)
NOAEL	: 6,000 ppm
Result	: positive

Carcinogenicity - Assessment	: Occurrence of tumors has been observed at excessive exposure levels. This can be considered as not relevant for possible carcinogenicity to humans during normal use.
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Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species	: Rat, male and female
Application Route	: inhalation (vapor)
Exposure time	: 12 month(s)
NOAEC	: 1.8 mg/l
Result	: negative
Remarks	: Based on data from similar materials

Carcinogenicity - Assessment	: Not classifiable as a human carcinogen.
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Reproductive toxicity

Not classified based on available information.

Based on available data, the classification criteria are not met.

Components:**malathion (ISO) [containing $\leq 0,03$ % isomalathion]:**

Effects on fertility	: Test Type: Two-generation study Species: Rat, male and female General Toxicity F1: NOAEL: 132 - 152 mg/kg bw/day Symptoms: Reduced offspring weight gain.
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Effects on fetal development	: Test Type: Embryo-fetal development Species: Rat General Toxicity Maternal: NOAEL: 400 mg/kg bw/day Teratogenicity: NOAEL: 800 mg/kg bw/day Result: No teratogenic effects.
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	: Test Type: Embryo-fetal development Species: Rabbit General Toxicity Maternal: NOAEL: 25 mg/kg bw/day Teratogenicity: NOAEL: 25 mg/kg bw/day Result: No teratogenic effects.
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Reproductive toxicity - Assessment	: Animal testing showed no reproductive toxicity.
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essment

acetic anhydride:

Effects on fetal development : Test Type: reproductive and developmental toxicity study
Species: Rabbit
Application Route: Oral
Dose: 2.5,16,74.3,345,1600mg/kgbw/d
Duration of Single Treatment: 13 d
General Toxicity Maternal: LOAEL: 74.3 mg/kg bw/day
Embryo-fetal toxicity.: NOAEL: 1,600 mg/kg bw/day
Symptoms: Malformations were observed.
Result: negative
Remarks: Based on data from similar materials

Reproductive toxicity - Assessment : Weight of evidence does not support classification for reproductive toxicity

STOT-single exposure

May cause drowsiness or dizziness.
May cause drowsiness or dizziness.

Product:

Assessment : May cause drowsiness or dizziness.

Components:

acetic anhydride:

Assessment : The substance or mixture is not classified as specific target organ toxicant, single exposure.

STOT-repeated exposure

Not classified based on available information.
Based on available data, the classification criteria are not met.

Repeated dose toxicity

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Species : Rat
LOAEL : 34.4 mg/kg
Application Route : Oral - feed
Exposure time : 90 d
Target Organs : Nervous system
Symptoms : cholinesterase inhibition

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Species : Rat, male and female
NOAEC : 0.9 - 1.8 mg/l
Application Route : inhalation (vapor)
Exposure time : 12 Months

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acetic anhydride:

Species	: Rat, male and female
LOAEC	: 25 ppm
Application Route	: Inhalation
Test atmosphere	: vapor
Exposure time	: 2 weeks
Dose	: 25, 100, 400 ppm

Aspiration toxicity

May be fatal if swallowed and enters airways.
May be fatal if swallowed and enters airways.

Product:

The substance or mixture is known to cause human aspiration toxicity hazards or has to be regarded as if it causes a human aspiration toxicity hazard.

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

The substance does not have properties associated with aspiration hazard potential.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

May be fatal if swallowed and enters airways.

acetic anhydride:

The substance or mixture causes concern owing to the assumption that it causes a human aspiration toxicity hazard.

Experience with human exposure

Components:

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Skin contact	: Symptoms: Repeated exposure may cause skin dryness or cracking.
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Further information

Product:

Remarks	: Symptoms of overexposure may be headache, dizziness, tiredness, nausea and vomiting. Concentrations substantially above the TLV value may cause narcotic effects. Solvents may degrease the skin.
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Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

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Remarks : The active ingredient malathion is a cholinesterase inhibitor of low mammalian toxicity. However, prolonged storage or storage at too high temperatures may induce formation of the much more toxic and synergistic contaminant isomalathion (LD50, oral, rat, 89 mg/kg). Both malathion and isomalathion rapidly enter the body on contact with all skin surfaces and eyes.
Repeated exposures to cholinesterase inhibitors such as malathion or isomalathion may, without warning, cause increased susceptibility to doses of any cholinesterase inhibitor.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Remarks : Vapour concentrations above recommended exposure levels are irritating to the eyes and the respiratory tract, may cause headaches and dizziness, are anaesthetic and may have other central nervous system effects. Prolonged and/or repeated skin contact with low viscosity materials may defat the skin resulting in possible irritation and dermatitis. Small amounts of liquid aspirated into the lungs during ingestion or from vomiting may cause chemical pneumonitis or pulmonary edema.

12. ECOLOGICAL INFORMATION

Ecotoxicity

Product:

Ecotoxicology Assessment

Acute aquatic toxicity : Very toxic to aquatic life.

Chronic aquatic toxicity : Very toxic to aquatic life with long lasting effects.

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): 0.18 mg/l
Exposure time: 96 h

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia similis (Water flea)): 1.71 µg/l
Exposure time: 48 h

Toxicity to algae/aquatic plants : IC50 (Selenastrum capricornutum (green algae)): 4.06 mg/l
Exposure time: 72 h

M-Factor (Acute aquatic toxicity) : 100

Toxicity to fish (Chronic toxicity) : NOEC (Oncorhynchus mykiss (rainbow trout)): 0.021 mg/l
Exposure time: 37 d

Toxicity to daphnia and other aquatic invertebrates (Chronic) : NOEC (Daphnia magna (Water flea)): 0.00006 mg/l
Exposure time: 21 d

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ic toxicity)

M-Factor (Chronic aquatic toxicity) : 1,000

Toxicity to soil dwelling organisms : (Eisenia fetida (earthworms)): 613 mg/kg
Exposure time: 14 d

Toxicity to terrestrial organisms : LD50 (Colinus virginianus (Bobwhite quail)): 359 mg/kg
Exposure time: 5 d

LC50 (Colinus virginianus (Bobwhite quail)): 3,497 mg/kg
Exposure time: 5 d
Remarks: Dietary

LD50 (Anas platyrhynchos (Mallard duck)): > 2,250 mg/kg

LD50 (Apis mellifera (bees)): 0.38 µg/bee
End point: Acute oral toxicity

Ecotoxicology Assessment

Toxicity Data on Soil : Harmful to the soil environment.

Other organisms relevant to the environment : Harmful to terrestrial vertebrates., Harmful to terrestrial invertebrates.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Toxicity to fish : LL50 (Oncorhynchus mykiss (rainbow trout)): 2 - 5 mg/l
Exposure time: 96 h
Method: OECD Test Guideline 203

Toxicity to daphnia and other aquatic invertebrates : EL50 (Daphnia magna (Water flea)): 1.4 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202

Toxicity to algae/aquatic plants : EL50 (Pseudokirchneriella subcapitata (green algae)): 1 - 3 mg/l
Exposure time: 24 h
Method: OECD Test Guideline 201

Toxicity to daphnia and other aquatic invertebrates (Chronic toxicity) : EL50 (Daphnia magna (Water flea)): 0.89 mg/l
Exposure time: 21 d
Method: OECD Test Guideline 211

Toxicity to microorganisms : LL50 (Tetrahymena pyriformis): 677.9 mg/l
Exposure time: 72 h
Test Type: Growth inhibition

acetic anhydride:

Toxicity to fish : LC50 (Oncorhynchus mykiss (rainbow trout)): > 300.82 mg/l
Exposure time: 96 h
Test Type: semi-static test

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Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 300.82 mg/l
Exposure time: 48 h
Method: OECD Test Guideline 202
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (Skeletonema costatum (marine diatom)): > 300.82 mg/l
Exposure time: 72 h
Test Type: static test
Remarks: Based on data from similar materials

EC50 (Skeletonema costatum (marine diatom)): 300.82 mg/l
Exposure time: 72 h
Test Type: static test
Remarks: Based on data from similar materials

Toxicity to microorganisms : NOEC (Pseudomonas putida): 1,150 mg/l
Exposure time: 16 h
Test Type: Growth inhibition

calcium bis(dodecylbenzenesulphonate), branched:

Toxicity to fish : LC50 (Fish): > 1 - 10 mg/l
Exposure time: 96 h
Remarks: Based on data from similar materials

Toxicity to daphnia and other aquatic invertebrates : EC50 (Daphnia magna (Water flea)): > 1 - 10 mg/l
Exposure time: 48 h
Remarks: Based on data from similar materials

Toxicity to algae/aquatic plants : EC50 (algae): > 1 - 10 mg/l
Exposure time: 72 h
Remarks: Based on data from similar materials

Persistence and degradability

Components:

malathion (ISO) [containing ≤ 0,03 % isomalathion]:

Biodegradability : Result: Not readily biodegradable.

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Biodegradability : Result: Readily biodegradable.
Biodegradation: 58.6 %
Exposure time: 28 d
Method: OECD Test Guideline 301F
Remarks: Based on data from similar materials

acetic anhydride:

Biodegradability : Inoculum: activated sludge
Result: Readily biodegradable.

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Biodegradation: 96 %
Exposure time: 20 d
Remarks: Based on data from similar materials

calcium bis(dodecylbenzenesulphonate), branched:

Biodegradability : Result: Readily biodegradable.

Bioaccumulative potential

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 95
Remarks: Bioaccumulation is unlikely.

Partition coefficient: n-octanol/water : log Pow: 2.75

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Bioaccumulation : Remarks: The product/substance has a potential to bioaccumulate.

Partition coefficient: n-octanol/water : log Pow: 3.72
Method: QSAR

acetic anhydride:

Bioaccumulation : Species: Fish
Bioconcentration factor (BCF): 3.16
Method: QSAR

Partition coefficient: n-octanol/water : log Pow: -0.577 (25 °C)
pH: 7
Method: QSAR

calcium bis(dodecylbenzenesulphonate), branched:

Bioaccumulation : Bioconcentration factor (BCF): 1
Remarks: Bioaccumulation is unlikely.

Mobility in soil

Components:

malathion (ISO) [containing $\leq 0,03$ % isomalathion]:

Distribution among environmental compartments : Remarks: medium mobility in soil

Solvent naphtha (petroleum), heavy arom.; Kerosine — unspecified:

Distribution among environmental compartments : Remarks: Expected to partition to sediment and wastewater solids. Moderately volatile.

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Other adverse effects

Product:

Additional ecological information : An environmental hazard cannot be excluded in the event of unprofessional handling or disposal.
Very toxic to aquatic life with long lasting effects.

13. DISPOSAL CONSIDERATIONS

Disposal methods

Waste from residues : The product should not be allowed to enter drains, water courses or the soil.
Do not contaminate ponds, waterways or ditches with chemical or used container.
Send to a licensed waste management company.

Contaminated packaging : Empty remaining contents.
Dispose of as unused product.
Do not re-use empty containers.
Do not burn, or use a cutting torch on, the empty drum.

14. TRANSPORT INFORMATION

International Regulations

UNRTDG

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.
(Malathion)
Class : 9
Packing group : III
Labels : 9
Environmentally hazardous : yes

IATA-DGR

UN/ID No. : UN 3082
Proper shipping name : Environmentally hazardous substance, liquid, n.o.s.
(Malathion)
Class : 9
Packing group : III
Labels : Miscellaneous
Packing instruction (cargo aircraft) : 964
Packing instruction (passenger aircraft) : 964

IMDG-Code

UN number : UN 3082
Proper shipping name : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S.

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(Malathion)

Class	: 9
Packing group	: III
Labels	: 9
EmS Code	: F-A, S-F
Marine pollutant	: yes

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Not applicable for product as supplied.

Special precautions for user

The transport classification(s) provided herein are for informational purposes only, and solely based upon the properties of the unpackaged material as it is described within this Safety Data Sheet. Transportation classifications may vary by mode of transportation, package sizes, and variations in regional or country regulations.

15. REGULATORY INFORMATION

National regulatory information

Regulations on Occupational Safety and Health Facilities	: applicable
Standards for the Storage, Cleanup, Handling and Disposal of Industrial Waste	: applicable
Regulations on Labelling and Hazard Communication of Hazardous Chemicals	: applicable
Rules on Road Traffic Safety	: applicable
Standards of Permissible Exposure Limits in Workplace	: applicable
Rules on the Prevention of Poisoning from Organic Solvents.	: applicable
Standard for the Control of Designated Hazardous and Dangerous Chemicals	: Not applicable
Establishment Standards and Safety Control Regulations for Manufacturing, Storing, Processing Public Hazardous Substances and Flammable Pressurized Gases Places	: Quantity subject to control
Toxic and Concerned Chemical Substances Control Act	
Toxic chemical substances	: Not applicable
Concerned chemical substances	: Not applicable
Regulations for Governing Designating and Handling of Priority Management Chemicals	: applicable

The ingredients of this product are reported in the following inventories:

TCSI	: On the inventory, or in compliance with the inventory
TSCA	: Product contains substance(s) not listed on TSCA inventory.
AIIC	: Not in compliance with the inventory
ENCS	: On the inventory, or in compliance with the inventory
ISHL	: On the inventory, or in compliance with the inventory

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KECI	: On the inventory, or in compliance with the inventory
PICCS	: On the inventory, or in compliance with the inventory
IECSC	: On the inventory, or in compliance with the inventory
NZIoC	: Not in compliance with the inventory
TECI	: Not in compliance with the inventory

16. OTHER INFORMATION

Responsible Department	: FMC Corporation
Contact person	: Bee Huang
Revision Date	: 2026/02/20
Date format	: yyyy/mm/dd

Full text of other abbreviations

ACGIH	: USA. ACGIH Threshold Limit Values (TLV)
TW OEL	: Standards of Permissible Exposure Limits in Workplace

ACGIH / TWA	: 8-hour, time-weighted average
ACGIH / STEL	: Short-term exposure limit
TW OEL / TWA	: 8-hour time weighted average
TW OEL / STEL	: time weighted average for short term exposure

AIIC - Australian Inventory of Industrial Chemicals; ANTT - National Agency for Transport by Land of Brazil; ASTM - American Society for the Testing of Materials; bw - Body weight; CMR - Carcinogen, Mutagen or Reproductive Toxicant; DIN - Standard of the German Institute for Standardisation; DSL - Domestic Substances List (Canada); ECx - Concentration associated with x% response; ELx - Loading rate associated with x% response; EmS - Emergency Schedule; ENCS - Existing and New Chemical Substances (Japan); ErCx - Concentration associated with x% growth rate response; ERG - Emergency Response Guide; GHS - Globally Harmonized System; GLP - Good Laboratory Practice; IARC - International Agency for Research on Cancer; IATA - International Air Transport Association; IBC - International Code for the Construction and Equipment of Ships carrying Dangerous Chemicals in Bulk; IC50 - Half maximal inhibitory concentration; ICAO - International Civil Aviation Organization; IECSC - Inventory of Existing Chemical Substances in China; IMDG - International Maritime Dangerous Goods; IMO - International Maritime Organization; ISHL - Industrial Safety and Health Law (Japan); ISO - International Organisation for Standardization; KECI - Korea Existing Chemicals Inventory; LC50 - Lethal Concentration to 50 % of a test population; LD50 - Lethal Dose to 50% of a test population (Median Lethal Dose); MARPOL - International Convention for the Prevention of Pollution from Ships; n.o.s. - Not Otherwise Specified; Nch - Chilean Norm; NO(A)EC - No Observed (Adverse) Effect Concentration; NO(A)EL - No Observed (Adverse) Effect Level; NOELR - No Observable Effect Loading Rate; NOM - Official Mexican Norm; NTP - National Toxicology Program; NZIoC - New Zealand Inventory of Chemicals; OECD - Organization for Economic Co-operation and Development; OPPTS - Office of Chemical Safety and Pollution Prevention; PBT - Persistent, Bioaccumulative and Toxic substance; PICCS - Philippines Inventory of Chemicals and Chemical Substances

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es; (Q)SAR - (Quantitative) Structure Activity Relationship; REACH - Regulation (EC) No 1907/2006 of the European Parliament and of the Council concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals; SADT - Self-Accelerating Decomposition Temperature; SDS - Safety Data Sheet; TCSI - Taiwan Chemical Substance Inventory; TDG - Transportation of Dangerous Goods; TECI - Thailand Existing Chemicals Inventory; TSCA - Toxic Substances Control Act (United States); UN - United Nations; UNRTDG - United Nations Recommendations on the Transport of Dangerous Goods; vPvB - Very Persistent and Very Bioaccumulative; WHMIS - Workplace Hazardous Materials Information System

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